

# **Fewer but Better? Revisiting the Role of Fertility Decline in Economic Growth**

Ricardo M. López Busch

May, 2026

## **Abstract**

This paper revisits the theoretical relationship between fertility decline and economic growth. To that end, we develop an endogenous growth model that combines fishing-out effects in innovation and a fixed cost of family formation. This setup identifies two economic drivers of lower fertility with asymmetric implications for long-run economic growth. When the cost of having an extra child increases, families substitute quality for quantity of children, resulting in faster economic growth. By contrast, when the fixed cost of having a family increases, fertility rates decline without a compensation of human capital, resulting in slower innovation and lower economic growth. The model thus predicts that fertility trends are not predictive of future economic growth.

JEL classification:

Keywords: Education, Fertility, Competition, Economic Growth

# 1 Introduction

For a long time, economists and other social scientists have viewed in falling fertility rates both a cause and a consequence of economic development (see Herzer et al. (2012)). A consequence because economic growth tends to be accompanied by increases in the cost of raising a child (see Bloom et al. (2024)). A cause because lower fertility rates go hand in hand with higher investments in human capital per child due to the trade-off between quantity and quality of children (see Doepke et al. (2023)). In line with this logic, some influential growth theories identify declining fertility as a key element in the transition to modern economic growth (e.g., Galor and Weil (2002)).

This favorable view of low fertility rates has recently come under scrutiny. Basic demographic forces imply that fertility shapes the age structure of the population. High fertility rates increase the share of young cohorts, while persistently low fertility leads to population aging, which is the predominant trend in developed economies. One economic consequence associated with aged societies is that they are less entrepreneurial and innovative (see Liang et al. (2018) and Maestas et al. (2023)). These findings are consistent with predictions from endogenous growth models featuring “fishing-out” effects (see Jones and Vollrath (2013)).<sup>1</sup> Both theory and evidence therefore indicate that lower fertility rates may depress long-run economic growth by limiting the supply of skilled workers, researchers, and entrepreneurs required to sustain innovation.

Taken together, the literature documents two countervailing effects of falling fertility rates on economic growth. On the one hand, lower fertility facilitates higher investments in human capital per child. On the other hand, reduced fertility weakens economic dynamism by slowing innovation and entrepreneurial activity. Recent empirical evidence for developed economies indicates that the latter effect seems to dominate (see Kotschy and Sunde (2018) and Aksoy et al. (2019)). This probably means that the empirical relationship between fertility decline and economic growth has flipped over the course of development. But why? Why is lower fertility sometimes positively related to economic growth in the data and sometimes negatively? Is the relationship between lower fertility and economic growth necessarily positive or negative all the time?

The central idea of this paper is that the empirical relationship between falling fertility and economic growth is inherently ambiguous. One potential reason that we explore here is that not all factors driving fertility rates down trigger the Beckerian trade-off between quantity and quality of children. From a macroeconomic perspective, one can think about the Beckerian trade-off as a sort of stabilizing mechanism that compensates lower fertility with higher human capital, so demographic aging is compensated by higher individual productivity. However, it is likely that some fertility drivers do not trigger the Beckerian mechanism, meaning that falling fertility rates end depressing economic

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<sup>1</sup>The fishing out effect means that existing ideas contribute to the production of new ones with diminishing returns, implying that long-run rate of innovation ultimately depends on the growth rate of the aggregate supply of human capital.

growth through aging. As a consequence, the empirical relationship between falling fertility and economic growth mutates over time as the different causes of lower fertility gain and loss strength.

To formalize this idea, we build an endogenous growth model that combines fishing out effects in innovation and a Beckerian trade-off subject to a fixed cost of family formation. This setup allows us to identify two distinct economic drivers of lower fertility with asymmetric implications for long-run growth. Changes in the cost of having an extra child trigger the Beckerian trade-off between the quantity and quality of children, generating the compensating increase in human capital described above. By contrast, changes in the fixed cost of having a family are fully absorbed through fertility without human capital compensation. In the first case, the decrease in fertility correlates with stronger economic growth because innovation remains stable while human capital accumulation per person accelerates. In the second case, economic growth slows down because the fertility decline chokes the supply of aggregate human capital needed to keep the innovation engine active and human capital accumulation does not mitigate.

The remainder of the paper is as follows. Section two presents the building blocks of the model, while the third section explores the long-run predictions of the model.

## 2 The Model

This section presents a growth model with endogenous innovation and fertility that will help us conduct the discussion of ideas. The model is set up in discrete time, with periods being denoted by  $t \in \mathbb{N}_+$ . During each period, the economy is comprised by families making decisions about consumption, fertility, and education. Individuals supply labor inelastically and use their wage income to purchase units of final output to consume. Firms in turn hire labor to produce final goods under an endogenously determined technology.

To make the analysis short and clear, we consider the simplest demographic structure possible. Individuals are grouped in two overlapped generations called children and adults, with all meaningful decisions being made by the adult generation. Families are mono-parental and homogeneous in preferences and endowments, allowing us to consider the case of a representative family. Regarding mortality, children reach the next period with certainty, whereas all adults die at the end of the period. The total number of adults, and thus families, is denoted by  $L_t$ , and its evolution is governed by the fertility rate of families  $b_t$ :

$$L_{t+1} = b_t L_t. \quad (1)$$

## 2.1 Production and Technological Progress

At first, one might be tempted to think that lower fertility rates should have a positive contribution on economic growth due to the trade-off between quantity and quality of children. According to this idea, encouraging families to have fewer children facilitates higher investments in child quality. At the macroeconomic scale, this involves a higher accumulation of human capital per person at the expense of lower population growth. Therefore, if labor exhibits non-increasing returns in production, one may conclude that income per person should benefit from fewer but better children. But although this reasoning seems neat and plausible, we shall show that it may not hold when innovation is included in the analysis.

To fix ideas, let us consider the case of an economy that produces a homogeneous final good  $Y_t$ , which is produced by combining units of efficient hours worked or human capital  $H_{Y,t}$  and units of intermediate inputs according to the following production function:

$$Y_t = \frac{1}{\beta} H_{Y,t}^{1-\beta} \int_{A_t} x_t(j)^\beta ds, \quad (2)$$

where  $x_t(j)$  denotes the quantity of variety  $j$  used in production,  $A_t$  indicates the total number of varieties, and  $1 > \beta > 0$  measures the return of each intermediate variety. This sector takes the prices of intermediate inputs  $p_t(j)$  as given, so profit maximization delivers the convenient isoelastic demand curves:

$$x_t(j) = H_{Y,t} \left( \frac{1}{p_t(j)} \right)^{\frac{1}{1-\beta}}. \quad (3)$$

Each variety  $j$  is produced by an intermediate firm that enjoys the exclusive rights to produce it, giving them the monopolistic power to choose the price. Intermediate producers face a marginal production cost equal to  $\phi$ , and the operative profit of this sector is given by the difference between revenue and production cost:

$$\pi_t(j) = p_t(j)x_t(j) - \phi x_t(j). \quad (4)$$

Intermediate producers pick the price  $p_t(j)$  that maximizes their operative profit (4) subject to the demand curve (3), resulting in the following pricing strategy:

$$p(j) = \frac{\phi}{\beta}. \quad (5)$$

Taking into account that the optimal pricing strategy is to set a fixed markup, and assuming that  $\phi = \beta$  to simplify notation, it turns out the demand of each variety is proportional to the total amount of human capital hired by the final sector:

$$x_t(j) = H_{Y,t}. \quad (6)$$

The demand curve of intermediate inputs, combined with the optimal pricing strategy, implies that intermediate firms have an operating profit proportional to the total demand of the variety:

$$\pi_t(j) = (1 - \beta)H_{Y,t}. \quad (7)$$

Since the demand of human capital in the final output sector  $H_{Y,t}$  determines the extend of the market of all varieties, it turns out that the marginal productivity of human capital in the that sector is given by the number of intermediate varieties. The wage rate consistent with free entry into final output production is then given by the number of varieties:

$$w_t = \frac{A_t}{\beta}. \quad (8)$$

In order to obtain the rights to exclusive production, intermediate producers need to pay a fee each period equal to  $v(j)$ , this is, they rent out the rights to produce for one period. In equilibrium, intermediate producers bid for a variety until the fee eats up all the operating profit. No more than that because it leads to losses, and no less because that involves net profit and some other intermediate producer would be willing to outbid. It is therefore the case that the value or rental price of a variety needs to equal the operating profit of the intermediate producer:

$$v_t(j) = \pi_t(j). \quad (9)$$

Finally, there is a entrepreneurial sector that hires units of human capital  $H_{R,t}$  to produce new varieties and pays the market wage rate equal to  $w_t$ . There is free entry into entrepreneurship, so the zero profit condition imposes that all the revenue from this sector goes to pay for wages:

$$v_t A_t = w_t H_{R,t}. \quad (10)$$

Taking into account equations (8) and (9), we have that the amount of researchers hired is a fixed proportion of the market size of each variety:

$$H_{R,t} = \beta(1 - \beta)H_{Y,t}. \quad (11)$$

Human capital is then combined with existing varieties (knowledge) to produce new varieties as follows:

$$A_{t+1} - A_t = (H_{R,t})^{1-\phi} A_t^\phi. \quad (12)$$

This law of technological progress is not new in the literature (e.g., Galor and Weil (2002)). It features the usual "fishing out effect", meaning that the economy needs a sustained expansion of the total supply of human capital to beat the decreasing returns of existing ideas ( $1 > \phi > 0$ ) and keep the engine of progress active (see Strulik et al. (2013)). This well-known prediction brings us to the heart of this paper. There is no guarantee that families choose a mix of quantity and quality of children that benefits innovation. In other words, it is perfectly possible to find a macroeconomic landscape that combines high levels of human capital per person, low fertility rates, and low innovation due to a disconnection between the Beckerian trade-off and economic growth.

## 2.2 Families

This subsection characterizes families' fertility and human capital decisions, showing how different sources of fertility decline generate asymmetric effects on the aggregate supply of human capital. The asymmetry arises because not all drivers of lower fertility trigger the Beckerian trade-off between the quantity and quality of children. When the variable cost of having a child rises, families substitute quality for quantity, leaving the aggregate supply of human capital broadly unchanged. By contrast, when the fixed cost of starting a family rises, families reduce fertility without adjusting child quality, which unambiguously reduces the aggregate supply of human capital in the future.

To formalize this mechanism, we consider a standard logarithmic utility function of the form:

$$U[c, b, h] = \alpha \ln c_t + (1 - \alpha) (\ln b_t + \mu \ln h_{t+1}) + \bar{U}, \quad (13)$$

where  $1 > \alpha > 0$  governs the relative importance of consumption compared to children,  $1 > \mu > 0$  captures the weight placed on child quality relative to quantity, and  $\bar{U}$  is the fixed utility derived from having a family.

To complete the optimization problem, we assume that families are endowed with one unit of time, which they supply inelastically to the labor market at a wage  $w_t h_t$ . Labor income is the sole source of income and is devoted entirely to purchasing units of the homogeneous final output. Importantly, starting a family involves a fixed cost  $\bar{c}_t$ , which captures large expenditures that are independent of the number of children, such as housing, a car, or other foundational investments in family formation. The resulting budget constraint is:

$$w_t h_t - \bar{c}_t = c_t + b_t (\epsilon w_t h_t + \psi_t h_{t+1}), \quad (14)$$

where  $\epsilon$  denotes the opportunity cost of having a child in terms of forgone income, and  $\psi_t$  is the cost per unit of human capital. The opportunity cost  $\epsilon$  captures factors such as gender wage gaps and paid parental leave policies that determine the income parents must forego to raise a child, which we assume to stay constant for the time being. Then,  $\psi_t$  represents the unit cost of human capital investments, such as teacher salaries and healthcare services. Although  $\psi_t$  is taken as given by the family, it is endogenous to the economy and determined by market wages and production technology to be specified later.

**Definition 1.** *The problem of the representative household is to choose consumption, fertility, and child quality,  $\Omega_t^* = \{c_t^*, b_t^*, h_{t+1}^*\}$ , to maximize the utility function (13) subject to the budget constraint (14), given endowments and prices  $\{w_t, h_t, \bar{c}_t, \epsilon, \psi_t\}$ .*

Lagrangian optimization establishes that the optimal choices of the household must

satisfy the following first-order conditions (FOCs):

$$\frac{\alpha}{c_t} = \lambda_t, \quad (15)$$

$$\frac{1 - \alpha}{b_t} = \lambda_t(\epsilon w_t h_t + \psi_t h_{t+1}), \quad (16)$$

$$\frac{(1 - \alpha)\mu}{h_{t+1}} = \lambda_t b_t \psi_t. \quad (17)$$

Equation (15) indicates that families allocate income to consumption until its marginal utility equals the shadow value of income  $\lambda_t$ . Equation (16) shows that the marginal utility of an additional child must equal its marginal cost in terms of foregone consumption. Similarly, equation (17) states that the marginal utility of child quality  $h_{t+1}^*$  must equal its marginal cost in terms of foregone consumption.

Crucially, neither the FOC for fertility nor the FOC for child quality is affected by the fixed cost  $\bar{c}_t$ . Consequently, the demand for child quality  $h_{t+1}^*$  is independent of the fixed cost of family formation and is determined solely by the opportunity cost of children  $\epsilon$  and the unit cost of child quality  $\psi_t$ :

$$h_{t+1}^* = \frac{\mu}{1 - \mu} \frac{\epsilon w_t h_t}{\psi_t}. \quad (18)$$

Since changes in  $\bar{c}_t$  do not affect child quality  $h_{t+1}^*$ , families adjust only their fertility rate  $b_t^*$  in response to such shocks. Combining the FOCs of consumption and fertility with the budget constraint yields the optimal number of children:

$$b_t^* = \frac{(1 - \alpha)(1 - \mu)(w_t h_t - \bar{c}_t)}{\epsilon w_t h_t}. \quad (19)$$

This expression reveals the two economic sources of fertility decline in this model. First, a higher opportunity cost of children  $\epsilon$  reduces fertility through the standard Beckerian channel, simultaneously triggering an increase in child quality  $h_{t+1}^*$ . Second, a higher fixed cost of family formation  $\bar{c}_t$  reduces fertility without affecting child quality  $h_{t+1}^*$ , since  $\bar{c}_t$  does not enter the margin between quantity and quality of children. These two sources of fertility decline have asymmetric implications for the aggregate supply of human capital  $H_{t+1}^*$ , defined as:

$$H_{t+1}^* = b_t^* h_{t+1}^* L_t^* = \frac{\mu(1 - \alpha)(w_t h_t - \bar{c}_t)}{\psi_t} L_t^*. \quad (20)$$

**Proposition 1.** *In the model with a fixed cost of family formation, increases in the unit cost of child quality  $\psi_t$  and the fixed cost of family formation  $\bar{c}_t$  reduce the aggregate supply of human capital  $H_{t+1}^*$ . By contrast, the opportunity cost of children  $\epsilon$  has no*

effect on the aggregate supply of human capital  $H_{t+1}^*$ .

Proof: Taking the derivative of  $H_{t+1}^*$  with respect to  $\psi_t$  yields:

$$\frac{\partial H_{t+1}^*}{\partial \psi_t} = -\frac{\mu(1-\alpha)(w_t h_t - \bar{c}_t)}{\psi_t^2} L_t^* < 0.$$

The derivative with respect to  $\bar{c}_t$  implies that:

$$\frac{\partial H_{t+1}^*}{\partial \bar{c}_t} = -\frac{\mu(1-\alpha)}{\psi_t} L_t^* < 0.$$

Finally, the derivative with respect to  $\epsilon$  establishes that:

$$\frac{\partial H_{t+1}^*}{\partial \epsilon} = 0. \quad \square$$

Proposition 1 indicates that changes in the opportunity cost of having a child do alter fertility but not the aggregate supply of human capital. The reason is that this driver of fertility triggers the Beckerian trade-off, so there is a compensation in terms of child quality. By contrast, changes in the fixed cost of having a family do affect both fertility and the aggregate supply of human capital because the Beckerian trade-off does not mediate. Similarly, increases in the cost of human capital do reduce the aggregate supply of human capital because those shocks are fully absorbed by child quality, leaving fertility unchanged.

### 2.3 Human Capital Sector

This subsection describes the production or formation of human capital. We assume that human capital is produced by teachers, doctors, and similar professionals, so that a fraction  $1 > \kappa_t > 0$  of the labor force is devoted to this sector. In equilibrium, the total demand for child quality by families must equal the total supply of human capital:

$$L_t^* b_t^* h_{t+1}^* = E \kappa_t^* h_t^* L_t^*, \quad (21)$$

where  $E > 0$  denotes the productivity of the human capital sector. From this equilibrium condition we can derive the evolution of the fraction of the labor force devoted to human capital production:

$$\kappa_t^* = \frac{b_t^* h_{t+1}^*}{E h_t^*}. \quad (22)$$

We also assume free entry into this sector, so that total revenue is entirely used to pay the wages of teachers, doctors, and similar professionals:

$$\psi_t^* L_t^* b_t^* h_{t+1}^* = \kappa_t^* h_t^* L_t^* w_t^* h_t^*. \quad (23)$$



Taking into account the equilibrium expression for  $\kappa_t^*$  in equation (22), we can solve equation (23) to obtain the unit cost of human capital  $\psi_t^*$  as a function of the wage rate and the productivity of this sector:

$$\psi_t^* = \frac{w_t^*}{E}. \quad (24)$$

From these building blocks we obtain that innovation raises the cost of child quality by driving up the wage per efficient hour worked of teachers, doctors, and similar professionals. Thus, the relative economic effort per unit of human capital  $\psi_t^*/w_t^*h_t^*$  sinks as human capital accumulates.

### 3 Equilibrium and Long-Run Implications

This section puts together all the pieces of the model to analyze the impact of families' fertility and child quality choices on innovation and economic growth over the long run. One way to derive clean comparative statics and long-run predictions is to assume that the fixed cost of having a family  $\bar{c}_t$  grows in line with income  $w_t h_t$ . This makes sense considering that this fixed cost encompasses things like the cost of housing, which tend co-move with income levels. Thus, for the time being, we will assume that this proportion stays constant over time and derive steady state predictions.

**Assumption 1.** *The fixed cost of family formation is proportional to income, that is,  $\bar{c}_t = \delta w_t h_t$ , where  $0 < \delta < 1$  is a constant parameter capturing the income share devoted to fixed family formation costs.*

Under Assumption 1, the ratio  $\bar{c}_t/w_t h_t = \delta$  is constant over time, which allows us to characterize the long-run equilibrium in closed form. Substituting into the optimal fertility rate (19), we have that the fertility rate in equilibrium remains stationary and determined by the cost of family formation and opportunity cost of child quantity:

$$b^* = \frac{(1 - \alpha)(1 - \mu)(1 - \delta)}{\epsilon}. \quad (25)$$

Similarly, substituting the equilibrium price of child quality (24) into the demand of child quality (18), we have that human capital per person grows at a constant rate over generations determined in this case the opportunity cost of child quantity and the efficiency of the human capital sector:

$$\gamma_h^* = \frac{h_{t+1}^*}{h_t^*} = \frac{\mu \epsilon E}{1 - \mu}. \quad (26)$$

The combination of a constant fertility rate and a sustained pace of human capital accumulation implies that the aggregate supply of human capital grows at a constant rate over time. Substituting the equilibrium price of child quality (24) and Assumption 1 into equation (20), the growth rate of the aggregate supply of human capital is given by:

$$\gamma_H^* = (1 - \alpha)(1 - \delta)\mu\epsilon E. \quad (27)$$

From this expression, we can establish the long-run comparative statics that motivate this paper. Comparing (25) and (27), one can see that fertility rates and the aggregate supply of human capital are not monotonically related. Those economic factors encompassed in  $\epsilon$  trigger the Beckerian trade-off, meaning that the aggregate supply of human capital remains constant because individual human capital compensates. By contrast, economic factors embedded in  $\delta$  do not trigger the trade-off and therefore affect the aggregate supply of human capital.

**Proposition 2.** *Under Assumption 1, changes in the opportunity cost of children  $\epsilon$  have no effect on the growth rate of the aggregate supply of human capital  $\gamma_H^*$ . By contrast, an increase in the income share of fixed family formation costs  $\delta$  reduces the growth rate of the aggregate supply of human capital  $\gamma_H^*$ .*

Proof: Taking the derivative of  $\gamma_H^*$  with respect to  $\epsilon$  yields:

$$\frac{\partial \gamma_H^*}{\partial \epsilon} = 0.$$

The derivative with respect to  $\delta$  implies that:

$$\frac{\partial \gamma_H^*}{\partial \delta} = -\mu(1 - \alpha)E < 0. \quad \square$$

Recall that the aggregate supply of human capital is split across the final output sector, the entrepreneurial sector, and the human capital sector:

$$H_t^* = H_{Y,t}^* + H_{R,t}^* + H_{E,t}^*. \quad (28)$$

Since the total demand for child quality grows at the constant rate, the fraction of the labor force devoted to the human capital sector remains constant over time:

$$\kappa^* = (1 - \alpha)(1 - \delta). \quad (29)$$

Moreover, from equation (11), the amount of human capital hired by the entrepreneurial sector is a fixed proportion of that hired by the final output sector. It follows that the aggregate supply of human capital is split across sectors in constant shares along the steady state path:

$$H_{Y,t}^* = \frac{1 - \kappa^*}{1 + \beta(1 - \beta)} H_t^* = \beta_Y H_t^*, \quad (30)$$

and

$$H_{R,t}^* = \frac{(1 - \kappa^*)\beta(1 - \beta)}{1 + \beta(1 - \beta)} H_t^* = \beta_R H_t^*. \quad (31)$$

This equilibrium property allows us to express the macroeconomic law of innovation as a function of the total supply of human capital:

$$\gamma_{A,t}^* = \frac{A_{t+1}^*}{A_t^*} = \left( \frac{\beta_R H_t^*}{A_t^*} \right)^{1-\phi} + 1. \quad (32)$$

When the stock of ideas is large relative to the aggregate supply of human capital, the pace of innovation slows down due to fishing-out effects. Conversely, when the stock of ideas is small relative to the supply of human capital, diminishing returns work in the opposite direction and accelerate innovation. The fishing-out effect therefore implies that the stock of ideas fluctuates around the total supply of human capital, converging to a balanced growth path in the long run.

**Proposition 3.** *Under Assumption 1, the rate of innovation  $\gamma_{A,t}^*$  converges to the growth rate of the aggregate supply of human capital  $\gamma_H^*$  over the long run, that is,  $\gamma_A^{ss} = \lim_{t \rightarrow \infty} \gamma_{A,t}^* = \gamma_H^*$ .*

Proof: See Appendix 1.

With Proposition 3, we can establish the main comparative statics of the paper. As shown in Proposition 2, the growth rate of the aggregate supply of human capital responds asymmetrically to the two sources of fertility decline. When fertility falls because the opportunity cost of children  $\epsilon$  rises, the quantity-quality trade-off ensures that child quality increases enough to fuel the engine on innovation at a constant speed. By contrast, when fertility falls because the burden of fixed family formation costs  $\delta$  rises, no such compensating adjustment in child quality takes place, choking the aggregate supply of human capital needed to fuel the engine of innovation.

**Corollary 1.** *Under Assumption 1, an increase in the opportunity cost of children  $\epsilon$  has no impact on the long-run rate of innovation  $\gamma_A^{ss}$ . By contrast, an increase in the fixed cost of family formation  $\delta$  reduces the long-run rate of innovation  $\gamma_A^{ss}$ .*

Proof: Since  $\gamma_A^{ss} = \gamma_H^*$  from Proposition 3, the result follows directly from Proposition 2:

$$\begin{aligned} \frac{\partial \gamma_A^{ss}}{\partial \epsilon} &= \frac{\partial \gamma_H^*}{\partial \epsilon} = 0, \\ \frac{\partial \gamma_A^{ss}}{\partial \delta} &= \frac{\partial \gamma_H^*}{\partial \delta} < 0. \quad \square \end{aligned}$$

Before moving on, notice that Corollary 1 makes a statement about the long-run and does not mention the transitional dynamics. The reason is that, in the short-run,

an increase in the fixed cost of having a family does encourage innovation to some degree. Recall that the associated decline in fertility reduces the total demand of human capital, and thus, the fraction of labor devoted to the human capital sector. In other words, there is more people available for research in the short-term if fertility declines without triggering the Beckerian trade-off. This implies that the transitional dynamics are obscured by the negative growth effect described in Corollary 1 and the positive level effect caused by the reallocation of human capital across sectors. The conclusion is that the impact of a higher  $\delta$  becomes perceptible only in the long-run.

Now we look into the dynamics of income per worker  $y_t^* = w_t^* h_t^*$ , which is the measure of economic growth in this model, and it is determined by the number of varieties and the stock of human capital per person:

$$y_t^* = w_t^* h_t^* = A_t^* h_t^*. \quad (33)$$

Computing gross growth rates and evaluating at the steady state, it follows that long-run economic growth is the product of innovation and human capital accumulation per person:

$$\gamma_y^{ss} = \gamma_A^{ss} \gamma_h^{ss}. \quad (34)$$

**Corollary 2.** *Under Assumption 1, an increase in the opportunity cost of children  $\epsilon$  raises long-run economic growth  $\gamma_y^{ss}$ . By contrast, an increase in the fixed cost of family formation  $\delta$  reduces long-run economic growth  $\gamma_y^{ss}$ .*

Proof: The proof follows directly from Corollary 1 and equation (26). Since  $\gamma_h^{ss}$  increases with  $\epsilon$  and is independent of  $\delta$ , and  $\gamma_A^{ss}$  increases with  $\epsilon$  and decreases with  $\delta$ , the results for  $\gamma_y^{ss} = \gamma_A^{ss} \cdot \gamma_h^{ss}$  follow immediately.  $\square$

Corollary 2 states the central message of this paper. Falling fertility rates and economic growth do not need to share a monotonic relationship in data, it depends on the source of fertility decline. To see this, consider two policies that reduce fertility through different channels. First, policies that reduce the opportunity cost of children  $\epsilon$ —such as closing gender wage gaps or abolishing child labor—lower fertility by making children relatively more expensive in terms of forgone income. Since these policies operate through  $\epsilon$ , they trigger the Beckerian quantity-quality trade-off and families respond by having fewer but better children, leaving the aggregate supply of human capital  $H_{t+1}^*$  unchanged. From Corollary 2, such policies are unambiguously growth enhancing, as higher human capital per person  $\gamma_h^{ss}$  raises long-run income per worker without depressing innovation.

Now, consider policies or structural trends that raise the fixed cost of family formation  $\delta$  such as rising housing prices or other large upfront costs associated with starting a family. These operate through  $\delta$  and reduce fertility without triggering any compensating increase in child quality. As a result, the aggregate supply of human capital contracts,

innovation slows, and long-run economic growth declines. The model therefore delivers a sharp and empirically relevant prediction: not all drivers of lower fertility have the same impact on economic growth, so fertility dynamics do not convey predictive information about future economic growth.

## Appendix 1: Proof of Proposition 3

From equation (31) we know that  $H_{R,t} = \beta_R H_t$ , where  $\beta_R$  is a constant. Since  $H_t$  grows at a constant rate equal to  $\gamma_H^* > 0$  from equation (27), it follows that  $H_{R,t}$  also grows at the constant rate  $\gamma_H^*$ . Now, define the ratio  $x_t = H_{R,t}/A_t$ , whose law of motion is given by:

$$\frac{x_{t+1}}{x_t} = \gamma_t^x = \frac{\gamma_H^*}{\gamma_A^*}.$$

Recall that  $\gamma_{A,t}^*$  is defined by equation (32), so we can substitute out to reach the dynamic equation:

$$\gamma_t^x = \frac{\gamma_H^*}{1 + x_t^{1-\phi}}.$$

Upon examination, it turns out that the derivative of this law of motion is negative  $\partial \gamma_t^x / \partial x_t < 0$ ,  $\forall x_t > 0$ . Moreover, this dynamic equation exhibits a unique steady state equal to:

$$x_{ss} = (\gamma_H^* - 1)^{\frac{1}{1-\phi}}.$$

Note that this steady state satisfies  $\gamma_t^x > 1$ ,  $\forall x_t < x_{ss}$  and  $\gamma_t^x < 1$ ,  $\forall x_t > x_{ss}$ , which combined with the sign of the derivative establishes the globally stable nature of the steady state  $x_{ss}$  for any initial condition  $x_0 > 0$ . It therefore holds that:

$$\lim_{t \rightarrow \infty} \gamma_t^x = 1,$$

which implies that the rate of technological progress must converge to  $\gamma_H^*$  over the long run:

$$\lim_{t \rightarrow \infty} \frac{\gamma_H^*}{\gamma_{A,t}^*} = 1. \quad \square$$

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